The Structural Polynomial of a Parity Sequence and Its Applications

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Abstract

We develop a polynomial representation for parity sequences of generalised Collatz-type maps (g, h, q), working in the ring $\mathbb{Z}[g, h]$ rather than specialising immediately to numerical values. The central object is the *structural polynomial* $k_w(g, h)$ attached to a parity sequence, which encodes the accumulated additive contribution of the odd steps independently of the parameter q.

The main results are: (1) a *path identity* $h^e b = g^o a + qk_w(g, h)$ valid for any parity sequence; (2) a *geometric polynomial theorem* showing that Steiner words $\mathrm{St}(\alpha, \beta)$ have $k(g, h) = (g^\alpha - h^\alpha)/(g - h)$, independent of β; (3) a *composition rule* $k_{Q\circ P} = g^{o_Q} k_P + h^{e_P} k_Q$; and (4) a *cycle element identity* $a = qk_w/(h^e - g^o)$ giving the fixed point as a rational function of (g, h, q).

Beyond these core results, we develop a system of five equivalent representations of any parity sequence — its bits (the p-value), the structural polynomial $k_w(g, h)$, its specialisation $k_w(3, 2)$, the σ-polynomial $\sigma_w(u, v)$, and the Collatz orbit ϕ_w — connected by explicitly invertible transformations. We also introduce a successor operation on k_w in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$ that generates all cycle representatives by rotation, and we warn that polynomial non-divisibility of $h^e - g^o \nmid k_w$ in $\mathbb{Z}[g, h]$ does not preclude pointwise divisibility at specific integer bases. Applications to the $2a + 1$ composition identity are developed in the companion paper [4].

Contents

1 Introduction

A *generalised Collatz-type map* with basis (g, h, q) consists of two elementary operations on the positive integers: an *odd step* $x \mapsto gx + q$ and an *even step* $x \mapsto x/h$. The standard Collatz map (the Syracuse odd-only iteration) uses basis $(g, h, q) = (3, 2, 1)$; the $5x + 1$ analogue uses $(5, 2, 1)$. A *parity sequence* is a finite word over $\{O, E\}$ specifying the order in which these steps are applied. Each parity sequence acts as an affine map on its domain of definition.

The central algebraic object we study is the *structural polynomial* $k(g, h) \in \mathbb{Z}[g, h]$ attached to a parity sequence. It captures the additive part of the parity sequence's affine action in a basis-independent form. The key advantage of working in $\mathbb{Z}[g, h]$ rather than substituting $g = 3$, $h = 2$ at the outset is that polynomial identities can be proved and, just as importantly, their *failure for other bases* can be identified precisely. A numerical check that an identity holds at $(3, 2)$ gives no information about why it fails at $(5, 2)$; working symbolically makes the obstruction explicit.

The power of the symbolic approach is illustrated dramatically by the $5x + 1$ system, which possesses a non-trivial cycle $17 \to 43 \to 27 \to 17$ under the Syracuse $5x+1$ map. Via the cycle element identity (Section 5), the fixed point $a = 17$ emerges as a rational function of (g, h, q): it equals $q \cdot k(g, h)/(h^e - g^o)$, where $k(5, 2) = 51$ and $h^7 - g^3 = 128 - 125 = 3$, giving $a = 51/3 = 17$. The near-miss $h^e - g^o = 3$ is the arithmetic miracle that makes the cycle possible, and it is visible as a polynomial phenomenon.

The paper is organised as follows. Section 2 establishes the path identity and defines the structural polynomial. Section 3 proves the geometric polynomial theorem for Steiner words. Section 4 derives the composition rule. Section 5 applies these tools to cycle element identification, and includes a remark cautioning that polynomial non-divisibility does not imply pointwise non-divisibility. Section 6 develops the five representations of a parity sequence — the p-value, the σ-polynomial, $k_w(g, h)$, $k_w(3, 2)$, and the Collatz orbit ϕ_w — and unifies them in a single diagram. Section 7 collects worked examples illustrating the theory.

2 The Structural Polynomial and Path Identity

Definition 2.1 (Basis and elementary maps)**.** A *basis* is a triple (g, h, q) of positive integers. The associated elementary maps are the *odd step* $O \colon x \mapsto gx + q$ and the *even step* $E \colon x \mapsto x/h$.

Definition 2.2 (Parity sequence)**.** A *parity sequence* is a finite word $w = w_1 w_2 \cdots w_n$ (possibly empty, $n \ge 0$) over the alphabet $\{O, E\}$. It defines an affine map Φ_w by composing the corresponding elementary maps from left to right: $\Phi_w = \Phi_{w_n} \circ \cdots \circ \Phi_{w_1}$ (the empty composition is the identity map). We write $o = o(w)$ for the number of occurrences of O in w and $e = e(w)$ for the number of occurrences of E.

Remark 2.3. Throughout this paper we assume $h \mid x$ whenever an E-step is applied, so that all maps are defined on their natural domain.

Definition 2.4 (κ-tuple)**.** Let w be a parity sequence with $o \ge 1$ odd steps and e even steps. Number the O-steps from 0 to $o-1$ in the order they appear in w. For $0 \le j \le o-1$, let κ_j denote the number of E-steps that appear in w *strictly before* the $(j+1)$-th O-step. The sequence $(\kappa_0, \kappa_1, \ldots, \kappa_{o-1})$ is the *κ-tuple* of w. It satisfies $0 = \kappa_0 \le \kappa_1 \le \cdots \le \kappa_{o-1} < e$. For example, $w = OEEOEOEE$ has $o = 3$ odd steps at positions $0, 3, 5$, preceded by 0, 2, 3 even steps respectively, so $\kappa = (0, 2, 3)$.

Remark 2.5. For the standard Collatz basis ($g = 3$, $q = 1$), a parity sequence corresponds to a legal orbit if and only if its κ-tuple is strictly increasing; the polynomial framework works unchanged for non-strict profiles (see Example 7.3).

Definition 2.6 (Structural polynomial)**.** The *structural polynomial* of a parity sequence w with κ-tuple $(\kappa_0, \ldots, \kappa_{o-1})$ is

$$k_w(g,h) \;=\; \sum_{j=0}^{o-1} g^{o-1-j}\, h^{\kappa_j} \;\in\; \mathbb{Z}[g,h].$$

When $o = 0$, we set $k_w = 0$.

We can now state and prove the central algebraic identity.

Theorem 2.7 (Path identity)**.** *Let (g, h, q) be a basis and w a parity sequence with o odd steps and e even steps, having structural polynomial $k_w(g, h)$. If $b = \Phi_w(a)$ then*

$$h^e \cdot b \;=\; g^o \cdot a \;+\; q \cdot k_w(g,h). \tag{1}$$

Proof. We proceed by induction on the length $n = |w|$ of the parity sequence.

Base cases. If $w = O$ (single odd step), then $e = 0$, $o = 1$, $b = ga+q$, and $k_w = g^0 h^0 = 1$. The identity reads $b = ga + q \cdot 1$, which holds. If $w = E$ (single even step), then $o = 0$, $e = 1$, $b = a/h$, $k_w = 0$, and the identity reads $h \cdot (a/h) = a$, which holds.

Inductive step. Suppose (1) holds for all parity sequences of length less than n, and write $w = w' \cdot \sigma$ where $\sigma \in \{O, E\}$ is the last step. Let $c = \Phi_{w'}(a)$ be the intermediate value, so $b = \Phi_\sigma(c)$. By the inductive hypothesis,

$$h^{e'} \cdot c = g^{o'} \cdot a + q \cdot k_{w'}(g,h), \tag{2}$$

where $o' = o(w')$ and $e' = e(w')$.

Case $\sigma = E$. Then $b = c/h$, $o = o'$, and $e = e' + 1$. Since $h^{e'+1}(c/h) = h^{e'} c = g^{o'} a + q k_{w'}$ by (2), and the E-step appended at the end does not precede any O-step, so the κ-tuple of w equals that of w', and therefore $k_w = k_{w'}$. The identity (1) holds.

Case $\sigma = O$. Then $b = gc + q$, so $o = o' + 1$ and $e = e'$. The new O-step is the last (index $j = o'$) in w. All $e' = e$ even steps in w precede it (they all belong to w'), so $\kappa_{o'} = e'$. The structural polynomial of w is

$$k_w(g, h) = g \cdot k_{w'}(g, h) + h^{e'},$$

because the exponents of the first o' terms each gain one extra factor of g, and the new final O-step contributes $g^0 h^{e'} = h^{e'}$. Now compute:

$$\begin{aligned} h^e \cdot b &= h^{e'} \cdot (gc + q) \\ &= g \cdot h^{e'} c + q \cdot h^{e'} \\ &= g\big(g^{o'} a + q \cdot k_{w'}(g, h)\big) + q \cdot h^{e'} \quad \text{(by (2))} \\ &= g^{o'+1} a + q\big(g \cdot k_{w'}(g, h) + h^{e'}\big) \\ &= g^o \cdot a + q \cdot k_w(g, h). \end{aligned}$$

□

Remark 2.8. Two features of the path identity deserve emphasis.

1. The polynomial $k_w(g, h)$ is independent of q: the parameter q appears only as a uniform scalar multiplier. Thus the additive structure of parity sequences is entirely captured by the polynomial ring $\mathbb{Z}[g, h]$.

2. The polynomial $k_w(g, h)$ is independent of trailing E-steps: appending E-steps at the end only scales h^e on the left-hand side of (1), leaving k_w unchanged. This means k_w alone does not uniquely identify a parity sequence; the p-value (Section 6) resolves the ambiguity by appending a sentinel bit that records the length n.

3 Steiner Words and the Geometric Polynomial Theorem

Definition 3.1 (Steiner word)**.** For integers $\alpha \geq 1$ and $\beta \geq 0$, the *Steiner word* $\mathrm{St}(\alpha, \beta)$ is the parity sequence

$$\mathrm{St}(\alpha, \beta) \;=\; \underbrace{(OE)^\alpha}_{\alpha \text{ copies of } OE} \; \underbrace{E^\beta}_{\beta \text{ extra } E\text{-steps}} .$$

It has $o = \alpha$ odd steps and $e = \alpha + \beta$ even steps.

Remark 3.2. Steiner words are the natural building blocks of Collatz orbits under the Syracuse (odd-only) map: each Syracuse step $S(a) = (3a + 1)/2^{v_2(3a+1)}$ corresponds to a Steiner word $\mathrm{St}(1, \beta)$ where $\beta = v_2(3a + 1) - 1 \geq 0$. A block of α consecutive applications of S before the next element divisible by an extra power of h corresponds to $\mathrm{St}(\alpha, \beta)$.

The κ-tuple of $\mathrm{St}(\alpha, \beta)$ has a particularly simple form.

Lemma 3.3 (κ-tuple of a Steiner word)**.** *The κ-tuple of* $\mathrm{St}(\alpha, \beta)$ *is* $\kappa_j = j$ *for* $0 \leq j \leq \alpha - 1$.

Proof. In $(OE)^\alpha E^\beta$, the j-th O-step (zero-indexed) appears at position $2j + 1$. Exactly j E-steps appear before it (one after each of the first j O-steps). Hence $\kappa_j = j$. □

Theorem 3.4 (Geometric polynomial theorem). *For any $\alpha \geq 1$ and $\beta \geq 0$,*

$$k_{\mathrm{St}(\alpha,\beta)}(g,h) \;=\; \frac{g^{\alpha}-h^{\alpha}}{g-h} \;=\; \sum_{j=0}^{\alpha-1} g^{\alpha-1-j}h^{j}. \tag{3}$$

In particular, $k_{\mathrm{St}(\alpha,\beta)}$ is independent of β.

Proof. Substituting $\kappa_j = j$ from Lemma 3.3 into Definition 2.6:

$$k_{\mathrm{St}(\alpha,\beta)}(g,h) = \sum_{j=0}^{\alpha-1} g^{\alpha-1-j}h^{j} = \frac{g^{\alpha}-h^{\alpha}}{g-h},$$

where the last equality is the standard geometric series identity. Independence of β follows from Remark 2.8. □

4 Composition of Structural Polynomials

If P and Q are parity sequences, the composition $Q \circ P$ denotes the parity sequence obtained by concatenating P then Q: first apply P, then apply Q to the result.

Theorem 4.1 (Composition rule). *Let P and Q be parity sequences with o_P, e_P and o_Q, e_Q odd and even steps respectively. Then*

$$k_{Q\circ P}(g,h) \;=\; g^{o_Q}\cdot k_P(g,h) \;+\; h^{e_P}\cdot k_Q(g,h). \tag{4}$$

Proof. Let $(\kappa_0^P, \ldots, \kappa_{o_P-1}^P)$ and $(\kappa_0^Q, \ldots, \kappa_{o_Q-1}^Q)$ be the κ-tuples of P and Q respectively.

In the concatenated sequence $Q \circ P$, the first o_P odd steps come from P and the last o_Q odd steps come from Q. The E-steps of P all precede the E-steps of Q.

For the j-th odd step of P ($0 \leq j < o_P$): the number of E-steps preceding it in $Q \circ P$ equals κ_j^P (no E-steps from Q are involved).

For the j-th odd step of Q ($0 \leq j < o_Q$): the number of E-steps preceding it in $Q \circ P$ is $e_P + \kappa_j^Q$ (all e_P steps of P plus the κ_j^Q steps of Q that precede it).

Therefore the structural polynomial of $Q \circ P$ is

$$\begin{aligned} k_{Q\circ P}(g,h) &= \sum_{j=0}^{o_P-1} g^{(o_P+o_Q-1)-j}\, h^{\kappa_j^P} + \sum_{j=0}^{o_Q-1} g^{o_Q-1-j}\, h^{e_P+\kappa_j^Q} \\ &= g^{o_Q} \sum_{j=0}^{o_P-1} g^{o_P-1-j}\, h^{\kappa_j^P} + h^{e_P} \sum_{j=0}^{o_Q-1} g^{o_Q-1-j}\, h^{\kappa_j^Q} \\ &= g^{o_Q}\cdot k_P(g,h) + h^{e_P}\cdot k_Q(g,h). \end{aligned}$$

□

Remark 4.2 (Interpretation). The two terms in (4) have natural interpretations. The factor g^{o_Q} on k_P accounts for the fact that after P has run, Q applies o_Q further odd multiplications, each of which scales the additive accumulation from P by g. The factor h^{e_P} on k_Q accounts for the fact that Q's own E-accumulation exponents are displaced upward by the e_P even divisions already consumed by P.

5 The Cycle Element Identity

Setting $a = b$ in the path identity (1) immediately gives the fixed-point equation for a parity sequence.

Theorem 5.1 (Cycle element identity)**.** *Let w be a parity sequence with o odd steps, e even steps, and structural polynomial $k_w(g,h)$. A positive integer a satisfies $\Phi_w(a) = a$ if and only if*

$$a \;=\; \frac{q \cdot k_w(g,h)}{h^e - g^o}. \tag{5}$$

Such a fixed point exists at basis (g,h,q) if and only if $h^e > g^o$ and $h^e - g^o$ divides $q \cdot k_w(g,h)$.

Proof. Setting $b = a$ in (1) gives $h^e a = g^o a + qk_w$, hence $a(h^e - g^o) = qk_w$, and (5) follows. For a to be a positive integer we need $h^e - g^o > 0$ (equivalently $h^e > g^o$) and divisibility. □

Remark 5.2. For path problems (no fixed-point condition), q is a free parameter of the basis and $k_w(g,h)$ encodes only the parity sequence's structural contribution. For cycle problems, q is no longer free: setting $d = h^e - g^o$, equation (5) requires $d \mid q \cdot k_w(g,h)$, so q must be a multiple of $d/\gcd(d, k_w(g,h))$. The smallest valid choice is exactly $q_w = d/\gcd(d, k_w(g,h))$ (Corollary 5.5); in particular, q must be a divisor of d. In particular, $q_w = 1$ if and only if $d \mid k_w(g,h)$ at the given integer basis (g,h).

Remark 5.3 (Polynomial vs. pointwise divisibility). It is tempting to argue about $(h^e - g^o) \mid k_w(g,h)$ at the polynomial level in $\mathbb{Z}[g,h]$. This does not work: polynomial non-divisibility does *not* imply pointwise non-divisibility. The $5x+1$ cycle at 17 (Example 7.1) is a concrete witness: the polynomial $h^7 - g^3$ does not divide $g^2 + gh + h^4$ in $\mathbb{Z}[g,h]$, yet $(2^7 - 5^3) = 3$ divides $k_w(5,2) = 51$ in $\mathbb{Z}$. The cycle exists because of the arithmetic accident $2^7 - 5^3 = 3$, which is invisible at the polynomial level. Any argument ruling out Collatz cycles must therefore work at the level of integer evaluations, not polynomial identities.

Proposition 5.4 (Successor on k_w around a cycle)**.** *Let $w = w_0 w_1 \cdots w_{n-1}$ be a cyclic parity sequence with o odd steps and e even steps in total, and let $w' = w_1 \cdots w_{n-1} w_0$ be the left rotation by one step. Then, working in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$,*

$$k_{w'}(g,h) \;\equiv\; \begin{cases} g \cdot k_w(g,h) & \text{if } k_w(1,0) = 1 \text{ (w begins with an O-step)}, \\ h^{-1} \cdot k_w(g,h) & \text{if } k_w(1,0) = 0 \text{ (w begins with an E-step)}. \end{cases}$$

In the O-step case, the unreduced product $g \cdot k_w(g,h)$ promotes the leading monomial $g^{o-1} \mapsto g^o$, and the natural recurrence

$$k_{w'}(g,h) \;=\; g \cdot k_w(g,h) + (h^e - g^o)$$

replaces this g^o by h^e — exactly the substitution $g^o \equiv h^e \pmod{h^e - g^o}$ — keeping the degree at $o - 1$ in g. The modulus $h^e - g^o$ is the cycle denominator from Theorem 5.1, so this operation is only natural in the context of a fixed cycle (o, e). As a consequence, the n rotations of w produce exactly n values of $k_{w'}(g,h)$ in $\mathbb{Z}/(h^e - g^o)\mathbb{Z}$, one for each cycle representative.

Corollary 5.5 (Canonical cycle basis)**.** *Let w be a parity sequence with o odd and e even steps, let $d = h^e - g^o > 0$, and assume* $\gcd(g,h) = 1$*. Set*

$$\delta \ := \ \gcd(d,\, k_w(g,h)), \qquad q_w \ := \ \frac{d}{\delta}, \qquad a_w \ := \ \frac{k_w(g,h)}{\delta}.$$

Then a_w is a positive integer, the basis (g,h,q_w) admits the integer cycle $\Phi_w(a_w) = a_w$, and q_w is the smallest positive integer q for which an integer fixed point exists at basis (g,h,q).

Proof. By construction $\delta \mid d$ and $\delta \mid k_w$, so $q_w \cdot k_w = (d/\delta) \cdot k_w = d \cdot (k_w/\delta)$, which is divisible by d. Hence $a_w = q_w \cdot k_w/d = k_w/\delta$ is a positive integer and satisfies (5). Minimality of q_w follows since any q for which $d \mid q \cdot k_w$ must satisfy $d/\delta \mid q$, so $q_w = d/\delta$ is the smallest such value. $\square$

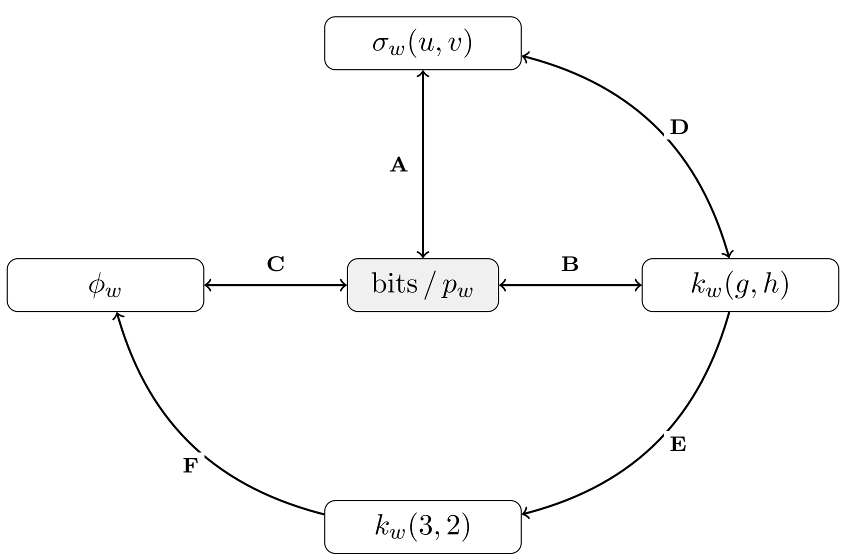
Remark 5.6. When $d \mid k_w$, the canonical basis has $q_w = 1$, recovering the standard Collatz setting. The coprimality hypothesis $\gcd(g,h) = 1$ is essential: when g and h share a common factor, additional congruence conditions on q arise beyond what $\gcd(d, k_w)$ captures. A worked example ($w = OEEOEOEE$, $q_w = 5$, $a_w = 29$) appears in Section 6.

6 Representations of a Parity Sequence

Definition 6.1 (Collatz orbit representation ϕ_w)**.** Let w be a parity sequence with o odd steps, e even steps, and $k_w(3,2) = k$. Set $q = 2^e - 3^o$ and define the sequence $a_0 = k$, $a_{i+1} = \Phi_{\text{step}}(a_i)$ under the $(3,2,q)$ Collatz map. The *Collatz orbit representation* of w is

$$\phi_w \ := \ (s_0, s_1, \ldots, s_{n-1}), \qquad s_i = \begin{cases} O & \text{if } a_i \text{ is odd,} \\ E & \text{if } a_i \text{ is even.} \end{cases}$$

Every parity sequence w of length n with o odd steps lives simultaneously in five equivalent representations. The canonical integer encoding (bits, p_w) sits at the centre; four structural representations — $\sigma_w(u,v)$, $k_w(g,h)$, $k_w(3,2)$, and ϕ_w (Definition 6.1) — surround it. The diagram below serves as a navigational frame for the subsections that follow; each subsection develops one or more nodes and the edges connecting them.



Edge	Pair	Forward	Inverse
A	bits $\leftrightarrow \sigma_w$	$\sigma_w(u,v) = \sum_{j=0}^{o-1} u^{o-1-j}\, v^{p_j}$	$p_w = \sigma_w(1,2) + 2^n$
B	bits $\leftrightarrow k_w(g,h)$	$k_w(g,h) = \sum_{j=0}^{o-1} g^{o-1-j} h^{\kappa_j}$	$p_w = 2^{o-1} k_w(\frac{1}{2}, 2) + 2^n$
C	bits $\leftrightarrow \phi_w$	$\phi_w[i] = O$ if $b_i = 1$, else $\phi_w[i] = E$	$b_i = 1$ if $\phi_w[i] = O$, else $b_i = 0$
D	$\sigma_w \leftrightarrow k_w(g,h)$	$k_w(g,h) = \sigma_w(gh,h)/h^{o-1}$	$\sigma_w(u,v) = k_w(u/v,\, v) \cdot v^{o-1}$
E	$k_w(g,h) \to k_w(3,2)$	$(g,h) \mapsto (3,2)$	(via **F** then **C** then **B**)
F	$k_w(3,2) \to \phi_w$ (Def. 6.1)	$q = 2^e - 3^o$, $a_0 = k_w(3,2)$; iterate $(3,2,q)$ map	(via **C** then **B**)

6.1 The p-value (edges B, C; centre node)

Since k_w is independent of trailing E-steps (Remark 2.8), it does not uniquely identify a parity sequence. A simple specialisation recovers a canonical integer encoding, and a sentinel bit resolves the trailing-E ambiguity.

Proposition 6.2 (Bit-encoding of parity sequences)**.** *Let $w = b_0 b_1 \cdots b_{n-1}$ be a parity sequence of length n, where $b_i = 1$ if the i-th step is an O-step and $b_i = 0$ if it is an E-step. Then*

$$N_w \;:=\; 2^{o-1} \cdot k_w\big(\tfrac{1}{2},\, 2\big) \;=\; \sum_{i=0}^{n-1} b_i\, 2^i,$$

the non-negative integer whose binary representation encodes w directly (LSB-first, $O = 1$, $E = 0$).

Proof. The j-th term of $k_w(g,h)$ is $g^{o-1-j} h^{\kappa_j}$. Evaluating at $g = 1/2$, $h = 2$ and multiplying by 2^{o-1}:

$$2^{o-1} \cdot \big(\tfrac{1}{2}\big)^{o-1-j} \cdot 2^{\kappa_j} = 2^{j+\kappa_j}.$$

Since κ_j counts E-steps before the j-th O-step and j counts O-steps before it, $j + \kappa_j = p_j$ is the position of the j-th O-step, i.e. the index i with $b_i = 1$. Summing over j gives $\sum_j 2^{p_j} = \sum_i b_i 2^i = N_w$. ☐

Sequences differing only in trailing E-steps (i.e. trailing $b_i = 0$) share the same N_w. To resolve this ambiguity we append a sentinel bit at position n.

Definition 6.3 (p-value)**.** The *p-value* of a parity sequence $w = b_0 b_1 \cdots b_{n-1}$ is

$$p_w \;:=\; \sum_{i=0}^{n-1} b_i\, 2^i \;+\; 2^n \;=\; 2^{o-1} \cdot k_w\big(\tfrac{1}{2}, 2\big) + 2^n.$$

Theorem 6.4 (p-value uniqueness)**.** *The map $w \mapsto p_w$ is a bijection from the set of all finite parity sequences over $\{O, E\}$ (including the empty sequence) to the set of integers ≥ 1. The empty sequence has $p_\varepsilon = 2^0 = 1$. For non-empty w, w is uniquely recovered from p_w by writing p_w in binary: the highest set bit (at position n) records the length, and bits $0, \ldots, n-1$ give the sequence with $1 = O$ and $0 = E$.*

Proof. The empty sequence has $n = 0$, $N_\varepsilon = 0$, and $p_\varepsilon = 0 + 2^0 = 1$. For non-empty w, given $p_w \geq 2$, let $n = \lfloor \log_2 p_w \rfloor \geq 1$ (the position of the highest set bit). Then bits $0, \ldots, n-1$ of p_w coincide with N_w and uniquely determine the positions of all O-steps, hence the full sequence w. The map is therefore injective. Surjectivity onto integers ≥ 1 follows since every positive integer has a unique highest set bit at some position $n \geq 0$: $n = 0$ gives the empty sequence, and $n \geq 1$ gives the parity sequence whose i-th step is O iff bit i of p_w is 1. □

Remark 6.5 (Successor as cyclic bit rotation). The successor $w \mapsto w'$ (Proposition 5.4) acts on the p-value as a *right* rotation of the lower n bits. Reading the data bits in the conventional MSB-first order as $b_{n-1}b_{n-2}\cdots b_1 b_0$, the right rotation moves the rightmost bit b_0 to the leftmost data position:

$$p_{w'} \;=\; 2^n + b_0\, 2^{n-1} + b_{n-1}\, 2^{n-2} + \cdots + b_1.$$

The sentinel bit at position n is unchanged, reflecting the fact that all cycle representatives have the same length n.

6.2 The σ-polynomial (edges A, D)

Change of variables

The substitution $g = 1/2$, $h = 2$ that produces N_w can be made polynomial (avoiding fractional exponents) by a change of variables.

Definition 6.6 (σ-polynomial)**.** For a parity sequence w of length n with o odd steps and O-step positions $p_0 < \cdots < p_{o-1}$, define

$$\sigma_w(u, v) \;:=\; k_w\big(\tfrac{u}{v},\, v\big) \cdot v^{o-1} \;=\; \sum_{j=0}^{o-1} u^{o-1-j}\, v^{p_j} \;\in\; \mathbb{Z}[u, v].$$

The exponent of u in the j-th term counts the O-steps *after* position j; the exponent of v is the bit-position of the j-th O-step. Two specialisations recover the objects already defined:

$$k_w(g, h) \;=\; \frac{\sigma_w(gh,\, h)}{h^{o-1}}, \qquad N_w \;=\; \sigma_w(1,\, 2),$$

and consequently $p_w = \sigma_w(1, 2) + 2^n$.

The successor operation on σ

Each step of the Collatz map acts on $\sigma_w(u, v)$ by an elegant polynomial operation [1]. Define

$$\gamma_w \;:=\; \begin{cases} 1 & \text{if } p_0 = 0 \text{ (i.e. } w \text{ begins with an } O\text{-step)}, \\ 0 & \text{otherwise.} \end{cases} \tag{γ}$$

Equivalently, $\gamma_w = 1$ if and only if the leading term $u^{o-1}v^{p_0}$ of σ_w has v-exponent zero. Define

$$\mathrm{succ}_\sigma(\sigma_w(u, v)) \;:=\; u^{\gamma_w} \cdot \sigma_w(u, v) \cdot v^{-1}.$$

Then $\mathrm{succ}_\sigma(\sigma_w(u, v)) = \sigma_{w'}(u, v)$, where w' is the parity sequence obtained by rotating w left by one step (moving the first bit to the last position).

Remark 6.7. Adopting the convention $0^0 = 1$, we have $\gamma_w = \sigma_w(1, 0)$: each term $1^{o-1-j} \cdot 0^{p_j}$ vanishes unless $p_j = 0$, which occurs precisely when $j = 0$ and w begins with an O-step.

The successor formula is valid in any ring where $u^o = 1$ and $v^n = 1$: the cycle closes after n steps since $v^{-n} = 1$, returning to σ_w. Complex roots of unity are one natural realisation of these constraints, but the algebraic structure depends only on these two relations.

Evaluation at roots of unity

Proposition 6.8. *Let $\omega_o = e^{2\pi i/o}$ and $\omega_n = e^{2\pi i/n}$ be primitive o-th and n-th roots of unity respectively. The evaluation*

$$\sigma_w(\omega_o, \omega_n) \;=\; \sum_{j=0}^{o-1} \omega_o^{o-1-j}\, \omega_n^{p_j}$$

expresses the parity sequence as a linear combination of n-th roots of unity, weighted by powers of ω_o. Each step of the cycle acts as

$$\mathrm{succ}_\sigma(\sigma_{w_k}(\omega_o, \omega_n)) \;=\; \omega_o^{\gamma_{w_k}} \cdot \sigma_{w_k}(\omega_o, \omega_n) \cdot \omega_n^{-1},$$

a multiplication by $\omega_o^{\gamma_{w_k}}$ (recording whether step k is an O-step) composed with a rotation ω_n^{-1} of the phase. Since $\omega_o^o = 1$ and $\omega_n^n = 1$, the cycle closes after exactly n steps. See [1] for the full development.

6.3 Worked example (all edges, $w = OEEOEOEE$)

The remaining nodes $k_w(3, 2)$ and ϕ_w are reached via edges E and F respectively; the return path from ϕ_w runs via **C** then **B** then **E**.

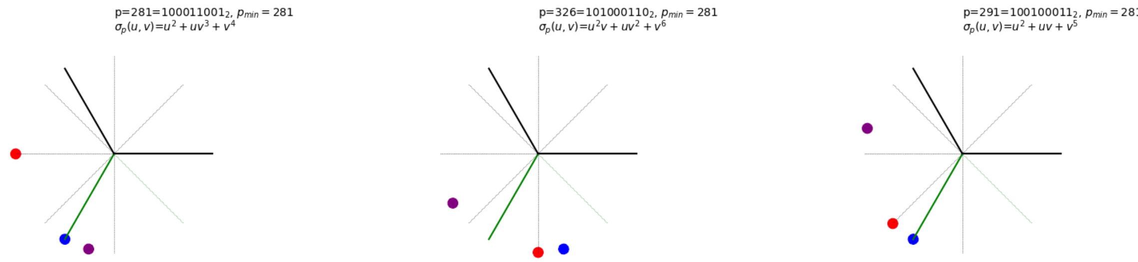


Figure 1: Three successive steps of the σ-successor for $w = OEEOEOEE$ ($n = 8$, $o = 3$), evaluated at (ω_3, ω_8). Coloured dots show the three monomials of σ_w; the green arrow is their sum (the value of σ_w as a complex number). *Left* (step 0, E-step, $\gamma = 0$): $\sigma = u^2 + uv^3 + v^4$. Notice all three dots rotate together by ω_8^{-1} — an E-step is a pure phase rotation with no change to the u-weights. *Centre* (after two E-steps): $\sigma = u^2v + uv^2 + v^6$ — the dots have rotated twice; the sum (green arrow) has changed direction. *Right* (after the first O-step, $\gamma = 1$): $\sigma = u^2 + uv + v^5$. The O-step multiplies by ω_3 in addition to rotating by ω_8^{-1}, redistributing the u-weights and visibly changing the triangle of dots.

C^{-1} $\quad b_8b_7\cdots b_0 = 1,0,0,1,0,1,0,0,1 \quad$ (O=1, E=0, b_8=1 sentinel)
$\quad$ O-positions p_0=0, p_1=3, p_2=5

A $\quad \sigma_w(u,v) = u^2 + u\,v^3 + v^5$

A^{-1} $\quad p_w = \sigma_w(1,2) + 2^8 = 41 + 256 = 297 = 100101001_2$

D $\quad k_w(g,h) = \sigma_w(gh,h)/h^2 = g^2 + gh^2 + h^3$

B^{-1} $\quad p_w = 2^{o-1}k_w\left(\frac{1}{2},2\right) + 2^8 = 41 + 256 = 297 = 100101001_2$ ✓

E $\quad k_w(3,2) = 3^2 + 3\cdot 2^2 + 2^3 = 29$

F $\quad q = 5,\ a_0 = 29$; iterate $(3,2,5)$:
$\quad 29 \xrightarrow{O} 92 \xrightarrow{E} 46 \xrightarrow{E} 23 \xrightarrow{O} 74 \xrightarrow{E} 37 \xrightarrow{O} 119 \xrightarrow{E} 62 \xrightarrow{E} 29 \;\Rightarrow\; \phi_w = OEEOEOEE$

C $\quad b_8b_7\cdots b_0 = 1,0,0,1,0,1,0,0,1 \quad$ (b_8=1 sentinel encodes n=8)

B $\quad \kappa_j = (0,2,3),\ \ k_w(g,h) = g^2 + gh^2 + h^3$ ✓

Since $\gcd(5,29) = 1$, the canonical basis gives $q_w = 5$, $a_w = 29$: no $3x$+1 cycle, but an integer $3x$+5 cycle at $a = 29$.

7 Examples

Example 7.1 (The $5x+1$ cycle at 17)**.** Under the Syracuse $5x+1$ map (basis $g = 5$, $h = 2$, $q = 1$), the odd integer 17 is a periodic point. Its orbit is

$$17 \xrightarrow{O} 86 \xrightarrow{E} 43 \xrightarrow{O} 216 \xrightarrow{E^3} 27 \xrightarrow{O} 136 \xrightarrow{E^3} 17.$$

The parity sequence is $w = OEOEEEOEEE$, with $o = 3$, $e = 7$.

The κ-tuple is $\kappa_0 = 0$, $\kappa_1 = 1$, $\kappa_2 = 4$ (the three O-steps are preceded by 0, 1, and 4 even steps respectively). Note that $(\kappa_0, \kappa_1, \kappa_2) = (0,1,4) \neq (0,1,2)$, so w is *not* a Steiner word and the geometric polynomial theorem does not apply directly. The structural polynomial is

$$k_w(g,h) = g^2h^0 + g^1h^1 + g^0h^4 = g^2 + gh + h^4.$$

Evaluating at $(g,h) = (5,2)$:

$$k_w(5,2) = 25 + 10 + 16 = 51, \qquad h^e - g^o = 2^7 - 5^3 = 128 - 125 = 3.$$

By Theorem 5.1,

$$a = \frac{1 \cdot 51}{3} = 17. \quad \checkmark$$

The near-miss $h^e - g^o = 3$ is what makes this cycle arithmetically possible: $2^7 - 5^3 = 3$ is a specific arithmetic accident with no analogue for the Collatz basis $(g,h) = (3,2)$. This also illustrates Remark 5.3: the polynomial $h^7 - g^3$ does not divide $g^2 + gh + h^4$ in $\mathbb{Z}[g,h]$, yet divisibility holds at $(5,2)$ because of this accident.

Example 7.2 (The $g = h^2 - 1$ family)**.** When divisibility *does* hold as a polynomial identity, it explains cycles across an entire family of bases at once. Consider $\mathrm{St}(1,1)^3$ (three copies of OEE), with $o = 3$, $e = 6$. By the composition rule:

$$\begin{aligned} k_{\mathrm{St}(1,1)^2}(g,h) &= g + h^2, \\ k_{\mathrm{St}(1,1)^3}(g,h) &= g^2 + gh^2 + h^4. \end{aligned}$$

On the locus $g = h^2 - 1$ one checks that $(g^2 + gh^2 + h^4) - (h^6 - g^3) = (g - h^2 + 1)(g^2 + gh^2 + h^4)$ vanishes identically on the locus $g = h^2 - 1$ (since $g - h^2 + 1 = 0$ there), so $k_{\mathrm{St}(1,1)^3} = h^6 - g^3$ and the cycle element $a = q \cdot k/(h^6 - g^3) = q$ for every such basis. At $q = 1$:

- $h = 2$, $g = 3$: orbit $1 \xrightarrow{O} 4 \xrightarrow{EE} 1$, three times.
- $h = 3$, $g = 8$: orbit $1 \xrightarrow{O} 9 \xrightarrow{EE} 1$, three times.

Example 7.3 (Non-Collatz parity sequence)**.** This example relaxes the requirement that the κ-tuple be strictly increasing. In the integer domain this corresponds to relaxing the rule that $gx + q$ is applied only to odd integers; in both settings the result is an "illegal" cycle that does not arise in standard Collatz dynamics.

Consider the parity sequence $w = OEEOOEEE$, with $o = 3$, $e = 5$. The κ-tuple is $\kappa_0 = 0$, $\kappa_1 = 2$, $\kappa_2 = 2$: both the second and third O-steps are preceded by exactly 2 even steps. The structural polynomial is

$$k_{OEEOOEEE}(g,h) = g^2h^0 + g^1h^2 + g^0h^2 = g^2 + h^2(g+1).$$

This w is *not* a valid standard Collatz parity sequence. The equal values $\kappa_1 = \kappa_2 = 2$ mean that two consecutive O-steps occur without any intervening E-step—the subsequence OO appears. Under standard Collatz, an O-step must be applied to an odd integer, so two consecutive O-steps are impossible: after $3n+1$ the result is even and must be halved before another O-step can occur. Equivalently, valid Collatz sequences require $\kappa_0 < \kappa_1 < \cdots < \kappa_{o-1}$ (strictly increasing).

If one relaxes this single condition—allowing $\kappa_j \le \kappa_{j+1}$—and correspondingly drives the $3x+1$ and $x/2$ operations by the letters of w rather than the parity of the current element, then cycles can exist. At $(g,h) = (3,2)$ we get $k_w(3,2) = 3^2 + 2^2(3+1) = 9 + 16 = 25$ and $h^e - g^o = 2^5 - 3^3 = 5$, so Theorem 5.1 yields cycle element $a = 25/5 = 5$. The orbit from $a = 5$ is

$$5 \xrightarrow{O} 16 \xrightarrow{E} 8 \xrightarrow{E} 4 \xrightarrow{O} 13 \xrightarrow{O} 40 \xrightarrow{E} 20 \xrightarrow{E} 10 \xrightarrow{E} 5,$$

which does return to 5. The single illegal step is $4 \xrightarrow{O} 13$: applying $3x+1$ to the even integer 4, forced by the repeated value $\kappa_1 = \kappa_2 = 2$.

Remark 7.4. The sequence $OEEOOEEE$ above has p-value $N_w + 2^8 = 25 + 256 = 281$: the integer 281 encodes the sequence directly in binary (with sentinel bit at position 8), and the glitched orbit was constructed by reading the bits of $281 - 256 = 25$ as a parity sequence.

8 Related Work

The structural polynomial $k_w(g,h)$ is closely related to the k-parameter appearing in the standard affine representation of Collatz orbits, where it appears as an entry in the affine transformation matrix. The Steiner word framework was developed in [2] as a natural decomposition of Syracuse orbits; the present paper provides its polynomial algebraic foundation.

The σ-polynomial $\sigma_w(u,v)$ and its successor operation at complex roots of unity were introduced in [1]; the present paper situates them within the broader five-representation framework and makes their connection to $k_w(g,h)$ via the change of variables $g = u/v$, $h = v$ explicit.

The composition rule (Theorem 4.1) and cycle element identity (Theorem 5.1) are applied in the companion paper [4] to prove a uniqueness theorem for the $2a+1$ Steiner composition identity and to characterise its domain constraint in terms of 2-adic valuations. Structural constraints on Collatz cycles derived from Steiner sentence analysis appear in [3].

References

[1] J. Seymour, *Exploring Cyclic Structures in Polynomial Rings with Applications to the Collatz Conjecture*, preprint, 2025. `https://wildducktheories.github.io/collatz/papers/14-uv-polynomial/uv-polynomial.pdf`

[2] J. Seymour, *Steiner Word Decomposition and Branch Length Distribution*, preprint, 2026. `https://wildducktheories.github.io/collatz/papers/67-branch-length-distribution/67-branch-length-distribution.pdf`

[3] J. Seymour, *Structural Constraints on Collatz Cycles via Steiner Sentence Analysis*, unpublished, 2026.

[4] J. Seymour, *The $2a+1$ Composition Identity: Uniqueness and Domain Constraint*, unpublished, 2026.